

HORTA, Portugal

WMO: 085060

Lat: 38.5297N

Lon: 28.6286W

Elev: 203

StdP: 14.59

Time Zone: -1.00 (AZR)

Period: 98-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	49.6	51.2	37.0	32.5	51.4	39.7	36.2	53.0	36.6	59.6	33.3	59.3	12.0	340	0.426

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	8.8	78.9	71.6	77.6	70.9	76.5	70.2	73.6	76.7	72.8	75.9	71.9	75.0	7.9	0

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
72.5	121.4	75.4	71.7	117.9	74.9	70.8	114.3	74.2	37.1	77.0	36.4	76.0	35.7	74.9	76.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 30.7	26.7	23.4	DB	46.1	82.1	2.2	1.6	44.6	83.3	43.3	84.3	42.1	85.2	40.5	86.3	(4)
(5)			WB	42.5	75.2	1.5	1.0	41.4	75.9	40.5	76.5	39.6	77.0	38.5	77.7	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	64.1	59.0	58.2	58.5	59.4	62.3	66.4	70.8	73.0	71.0	66.9	62.9	59.9	(6)
(7)		DBStd	5.87	2.64	2.73	2.63	2.53	2.35	2.77	2.31	2.01	2.72	3.43	3.38	3.36	(7)
(8)		HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)
(9)		HDD65	1106	185	191	201	167	87	17	0	0	1	20	79	158	(9)
(10)		CDD50	5138	280	230	264	283	382	492	644	715	631	523	386	308	(10)
(11)		CDD65	770	0	0	0	0	5	59	179	250	182	78	15	2	(11)
(12)		CDH74	1074	0	0	0	0	0	20	204	588	238	24	0	0	(12)
(13)		CDH80	12	0	0	0	0	0	0	1	9	2	0	0	0	(13)
(14)	Wind	WSAvg	10.7	12.2	11.9	12.5	11.4	10.3	9.4	8.3	8.1	9.2	11.4	11.1	12.7	(14)
(15)	Precipitation	PrecAvg	38.6	4.3	3.7	3.6	2.6	2.2	1.9	1.3	2.1	3.4	4.6	4.4	4.6	(15)
(16)		PrecMax	57.0	9.7	6.9	8.8	5.9	6.0	4.8	2.7	8.1	9.8	11.4	8.9	10.9	(16)
(17)		PrecMin	30.6	0.6	0.8	0.6	0.3	0.0	0.4	0.0	0.1	0.7	0.9	0.3	0.9	(17)
(18)		PrecStd	5.5	2.1	1.6	2.1	1.5	1.4	1.1	0.7	1.5	2.4	2.2	2.3	2.2	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	66.4	65.2	65.7	67.8	70.5	76.0	79.2	81.0	79.8	76.3	72.0	67.8	(19)
(20)			MCWB	60.5	60.6	61.1	60.9	64.9	69.9	71.6	72.4	72.5	69.5	67.1	64.3	(20)
(21)		2%	DB	64.4	63.8	64.1	65.9	68.8	73.8	77.4	79.4	77.9	74.4	69.9	66.4	(21)
(22)			MCWB	60.9	59.8	59.9	60.5	63.7	68.0	70.6	71.8	71.4	68.7	65.3	63.3	(22)
(23)		5%	DB	63.6	62.8	63.0	64.5	67.5	72.1	76.2	78.2	76.6	72.8	68.4	65.4	(23)
(24)			MCWB	60.4	59.3	59.3	59.5	62.8	66.5	69.8	71.2	70.3	68.0	64.4	62.5	(24)
(25)		10%	DB	62.8	62.0	62.1	63.3	66.3	70.8	74.9	77.1	75.3	71.6	67.4	64.6	(25)
(26)			MCWB	60.0	58.8	58.8	58.9	62.0	65.8	69.1	70.7	69.4	67.5	63.7	61.7	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	63.6	62.2	62.6	63.0	66.6	71.7	73.8	75.0	74.4	71.6	68.7	65.6	(27)
(28)			MCDWB	64.4	63.4	64.4	65.1	68.8	75.3	76.9	77.9	77.8	73.9	70.5	66.7	(28)
(29)		2%	WB	62.2	61.4	61.6	62.1	65.4	69.9	72.7	73.8	73.1	70.3	66.6	64.6	(29)
(30)			MCDWB	63.3	62.6	63.1	63.9	67.5	72.2	75.6	76.9	76.2	72.5	68.4	65.9	(30)
(31)		5%	WB	61.6	60.7	60.9	61.3	64.4	68.4	71.7	73.1	72.2	69.4	65.7	63.6	(31)
(32)			MCDWB	62.9	62.0	62.2	63.3	66.5	70.7	74.4	76.3	75.1	71.7	67.6	65.0	(32)
(33)		10%	WB	60.9	59.9	60.2	60.5	63.3	67.3	70.7	72.3	71.1	68.5	64.8	62.6	(33)
(34)			MCDWB	62.4	61.6	61.7	62.7	65.5	69.7	73.5	75.6	74.2	70.9	67.0	64.1	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	6.3	6.6	6.7	7.5	7.5	8.0	8.5	8.8	8.1	7.1	6.2	6.5	(35)
(36)			MCDBR	7.1	6.7	6.7	8.2	8.3	9.3	9.4	9.7	8.9	7.6	6.4	6.6	(36)
(37)			MCWBR	6.1	5.8	5.5	6.0	5.8	6.0	5.6	5.7	5.7	5.4	5.0	6.0	(37)
(38)		5% WB	MCDWB	5.4	6.3	5.8	7.0	7.2	8.1	8.0	8.5	7.7	6.7	5.8	5.7	(38)
(39)			MCWBR	5.9	6.2	5.4	5.8	5.6	5.7	5.7	5.7	5.8	5.7	5.4	6.1	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.354	0.361	0.391	0.384	0.379	0.378	0.373	0.368	0.366	0.368	0.354	0.355	(40)	
(41)		taud	2.506	2.486	2.388	2.432	2.475	2.511	2.536	2.573	2.579	2.554	2.539	2.482	(41)	
(42)		Ebn at Noon	260	274	275	284	286	285	285	285	280	268	258	250	(42)	
(43)		Edh at Noon	25	29	35	36	35	34	32	30	29	27	24	24	(43)	
(44)	All-Sky Solar Radiation	RadAvg	636	903	1257	1636	1884	2007	2069	1952	1528	1064	716	547	(44)	
(45)		RadStd	50	47	88	147	146	133	113	79	77	68	41	40	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (3 neighbors)		N/A	N/A	+1.84	N/A	+0.63	+0.94	N/A	N/A	N/A	N/A

Nomenclature: See separate page